

ARJUN YADAPADITHAYA

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PROFESSIONAL EXPERIENCE

Research Engineer

Illinois Institute of Technology, Chicago, IL

January 2024 – Apr 2026

- Driven the full lifecycle of schematic capture, component selection (BOM), and custom PCBs assembled a low-cost steam trap monitoring system, integrating thermocouples, piezoelectric accelerometers, and an ATmega based microcontroller.
- Led board bring-up and triaged complex hardware-software integration issues, utilizing oscilloscopes and logic analyzers to debug SPI and I2C communication bottlenecks.
- Automated data collection and logging with InfluxDB and Grafana to monitor system nodes, creating analysis tools that improved overall network stability and hardware reliability.

Senior Software Engineer

Capgemini, Bengaluru, India

August 2021 - August 2023

- Collaborated with cross-functional engineering teams to provide technical solutions for system inefficiencies, streamlining backend data structures to improve speed and reduce memory usage, addressed 2 PRs daily, worked with cross-functional teams and introduced features to enhance functionality and user experience.
- Transitioned deployment processes from Jenkins to Rancher for efficient container management, supporting virtualization and enabling scalability across multiple systems in a hybrid environment.

PROJECTS

Architectural Exploration of Dynamic Branch Predictors

January 2025 – April 2025

- Designed and coded three correlated dynamic branch predictor microarchitectures gpredict, gselect, gshare within strict hardware constraints 4-bit GHR, 16-entry BHT to optimize instruction-level parallelism.
- Conducted architectural trade-off analysis between aliasing and history depth; demonstrated how gselect indexing reduced collision mispredictions by segregating inner/outer loop branches in distinct BHT banks.
- Developed test automation infrastructure using a universal Verilog testbench and custom Python scripting to generate 6,000-line traces, verifying correct counter saturation and prediction timing via ModelSim waveform analysis.

Smart Energy Meter

March 2024 – Present

- Designed an analog and digital data-sniffing interface using the LM2902 op-amp to perform analog signal conditioning, converting raw sensor data into digital telemetry via linear regression.
- Implemented an ESP32-based solution to log data via Wi-Fi, transferring energy consumption data to a database for continuous monitoring, analysis, troubleshooting and insights on Grafana.

Battery Operated Radiator Control

October 2023 - Present

- Architected C++ firmware to drive a mechanical actuator over a precise 270-degree range for automated radiator control and implemented robust packet recovery for reliable RFM69HCW sub-GHz communication.
- Utilized Watchdog Timer (WDT) interrupts to manage deep-sleep cycles, significantly reducing active-duty cycle and extending battery life while maintaining periodic telemetry uploads to InfluxDB.
- Led system validation and real-world testing in a high-traffic school environment, optimizing hardware for vibration tolerance and signal reliability in a rugged setting.
- Achieved 74% cost savings in energy consumption and secured \$30,000 in funding as the Winner of the Kaplan Pitch Tank 2024, proving the hardware design's technical feasibility and market potential.

EDUCATION

Michigan Technological University, Houghton, MI

May 2029

Ph.D, Electrical and Computer Engineering

- Courses: Machine Learning, Advanced Engineering Mathematics, Prob & Stoch Processes.

Illinois Institute of Technology, Chicago, IL

May 2025

MS, Electrical and Computer Engineering, GPA: 3.9 / 4.0

- Courses: Digital Signal Processing, OOPS and ML with C++/Python, Hardware Software Co-design, Computer Vision & Image Processing, Hardware Security and Advanced Computer Architectures, Cloud Computing & Native Systems, Elements of Smart Grid.

SKILLS

- **Languages:** C, C++, Python, SQL, MATLAB, Verilog.
- **Technologies & Tools:** Linux, FreeRTOS, LTSpice, KiCAD, EAGLE, 3D CAD, PCBA, Sigma Studio, CCS, Git, PyTorch, TortoiseSVN, Grafana, Flask, MongoDB, Jenkins, Rancher, Angular, Raspberry Pi, Wi-Fi, Bluetooth, SPI, UART, I2C, PWM.